12401067.

Department of Computer Science and Engineering Punjabi University, Patiala
B. Tech 1st Year (1st Semester) CSE : Subject : Applied Physics
MST-I

Time: 1 Hour | **M. Marks: 15**

1. (a) Define quality factor in case of damped harmonic oscillator.
   (b) Write the condition for light damping in damped mechanical oscillator.
   (c) What is relation between damping factor and relaxation time?
   (d) How does the interference pattern by reflection in thin films differ from that by transmission? Why?
   (e) Distinguish between interference and diffraction. (5×1=5)
2. (a) A mass of 1 kg is attached to a spring of stiffness constant 16 Nm^{-1}. It oscillates in a viscous medium offering resistance proportional to velocity. If the energy of the system decays to 1/e of its initial value in 5 seconds, calculate damping constant and the Q-factor.
   (b) Discuss the interference in thin films from reflected light. (2, 3)
3. Derive the equation for simple harmonic motion for mechanical oscillator and find the expression for total energy of the oscillator.
   or
   Discuss the experimental setup in Michelson's interferometer. Discuss the formation of circular fringes. How is it useful in finding the wavelength of the light? (5)

$\frac{1}{2} A^2 \omega_0$

Tarun.

B. Tech. I CSE (SEM I) MST I Applied Physics

Time: 1 Hour **Max. Marks: 15**

Q1. (a) What are coherent sources of light?

(b) What are Lissajous figures?

(c) what is the relation between logarithmic decrement and relaxation time.

(d) Write the refractive index of Canada balsam material used in Nicol prism.

(e) Write the expression for thickness of anti-reflecting film? (5×1)

Q2. (a) Calculate the frequency and quality factor of a LCR circuit with L=2 mH, C=5 μF and R=0.2 ohm.

(b) Discuss the interference in thin films from reflected light. (2,3)

Q3. Derive the equation of motion for damped mechanical oscillator and discuss the case of critical damping.

or

Describe the Michelson interferometer and discuss the formation of circular fringes in it. How is it useful in finding the wavelength of the light? (5)

12 10 1078
Prem Kumar

Department of Basic and Applied Sciences

MST I (APPLIED PHYSICS-I, BAS-101) B. Tech-1 Semester-1

Time Allowed: **1 hour**

Maximum marks: **15**

Please mention your Group at the Top of answer sheet.

NOTE: All questions are compulsory.

Q1. (i) Are all periodic motions simple harmonic? Is the reverse true?
(ii) What do you mean by quality factor? Write its expression.
(iii) Explain why coherent sources are necessary for interference.
(iv) Why thin films appear coloured in white light?
(v) What should be the minimum thickness of a non-reflecting thin film? (1×5)

Q2. (a) What is simple harmonic motion? Show that the total energy of simple harmonic motion is constant.
(b) Calculate the thickness of a soap film (refractive index = 1.463) that will result in constructive interference of the reflected light, if the film is illuminated normally with light of wavelength 6000 Å. **(3, 2)**

Q3. What do you mean by damped oscillator? Derive its differential equation of motion and discuss the case of lightly damped system.

OR

Discuss principle, construction and working of Michelson's interferometer. Describe the formation of circular fringes. **(5)**